# Logger and Calculation Engine

1.0

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# Chapter 1

# Student Calculation Project

## 1.1 Introduction

This small multi-file C++ project demonstrates Doxygen documentation for a Logger class and a CalculationEngine class.

The Logger reports messages, while CalculationEngine performs a power calculation and uses Logger to report the operation.

# Chapter 2

# Todo List

**Member CalculationEngine::power (double base, int exponent) const**

Add validation for unsupported numeric input.

# Chapter 3

# Class Index

## 3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:


CalculationEngine
: Performs basic mathematical calculations . . . . . 9

Logger
: Provides simple application logging . . . . . 11

# Chapter 4

# File Index

## 4.1 File List

Here is a list of all files with brief descriptions:


CalculationEngine.h
: Declares mathematical operations provided by CalculationEngine . . . . . 13

Logger.h
: Declares the Logger class and log levels . . . . . 14

main.cpp . . . . . 14

# Chapter 5

# Class Documentation

## 5.1 CalculationEngine Class Reference

Performs basic mathematical calculations.

`#include <CalculationEngine.h>`

#### Public Member Functions

- CalculationEngine (const Logger &logger)
  *Creates a calculation engine.*
- double power (double base, int exponent) const
  *Calculates a base raised to an exponent.*

#### Private Attributes

- const Logger & logger

### 5.1.1 Detailed Description

Performs basic mathematical calculations.

The class uses a Logger object to report calculation activity.

### 5.1.2 Constructor & Destructor Documentation

#### 5.1.2.1 CalculationEngine()

```
CalculationEngine::CalculationEngine (
            const Logger & logger)  [inline], [explicit]
```

Creates a calculation engine.

**Parameters**

| | |
|---|---|
| *logger* | Logger used to report operations. |

### 5.1.3 Member Function Documentation

#### 5.1.3.1 power()

```
double CalculationEngine::power (
            double base,
            int exponent) const
```

Calculates a base raised to an exponent.

**Parameters**

| | |
|---|---|
| *base* | The number to be raised to a power. |
| *exponent* | The exponent. |

**Returns**

The calculated power.

**Todo** Add validation for unsupported numeric input.

### 5.1.4 Member Data Documentation

#### 5.1.4.1 logger

```
const Logger& CalculationEngine::logger  [private]
```

The documentation for this class was generated from the following files:

- CalculationEngine.h
- main.cpp

## 5.2 Logger Class Reference

Provides simple application logging.

`#include <Logger.h>`

### Public Types

- enum LogLevel { DEBUG , INFO , WARNING , ERROR }

  *Represents the severity of a log message.*

### Public Member Functions

- Logger (LogLevel level=INFO)

  *Constructs a Logger.*
- void log (LogLevel level, const char ∗message) const

  *Writes a message to the console.*

### Private Attributes

- LogLevel minimumLevel

### 5.2.1 Detailed Description

Provides simple application logging.

The Logger class writes messages with an associated log level.

### 5.2.2 Member Enumeration Documentation

#### 5.2.2.1 LogLevel

`enum Logger::LogLevel`

Represents the severity of a log message.

**Enumerator**

| | |
|---|---|
| DEBUG | Detailed diagnostic information. |
| INFO | General information about program execution. |
| WARNING | A condition that may require attention. |
| ERROR | An operation failed or encountered an error. |

### 5.2.3 Constructor & Destructor Documentation

#### 5.2.3.1 Logger()

```
Logger::Logger (
            LogLevel level = INFO)  [inline], [explicit]
```

Constructs a Logger.

**Parameters**

| | |
|---|---|
| *level* | Minimum log level to display. |

### 5.2.4 Member Function Documentation

#### 5.2.4.1 log()

```
void Logger::log (
            LogLevel level,
            const char * message) const
```

Writes a message to the console.

**Parameters**

| | |
|---|---|
| *level* | Severity of the message. |
| *message* | Text to be logged. |

### 5.2.5 Member Data Documentation

#### 5.2.5.1 minimumLevel

```
LogLevel Logger::minimumLevel  [private]
```

The documentation for this class was generated from the following files:

- Logger.h
- main.cpp

# Chapter 6

# File Documentation

## 6.1 CalculationEngine.h File Reference

Declares mathematical operations provided by CalculationEngine.

```
#include "Logger.h"
```

### Classes

- class CalculationEngine

  *Performs basic mathematical calculations.*

### 6.1.1 Detailed Description

Declares mathematical operations provided by CalculationEngine.

## 6.2 CalculationEngine.h

Go to the documentation of this file.

```
00001
00005
00006 #ifndef CALCULATION_ENGINE_H
00007 #define CALCULATION_ENGINE_H
00008
00009 #include "Logger.h"
00010
00016 class CalculationEngine {
00017 public:
00022     explicit CalculationEngine(const Logger& logger) : logger(logger) {}
00023
00031     double power(double base, int exponent) const;
00032
00033 private:
00034     const Logger& logger;
00035 };
00036
00037 #endif
```

## 6.3 Logger.h File Reference

Declares the Logger class and log levels.

### Classes

- class Logger

  *Provides simple application logging.*

### 6.3.1 Detailed Description

Declares the Logger class and log levels.

This file contains the logging interface used by the calculation engine in the practical project.

## 6.4 Logger.h

Go to the documentation of this file.

```
00001
00008
00009 #ifndef LOGGER_H
00010 #define LOGGER_H
00011
00018 class Logger {
00019 public:
00024     enum LogLevel {
00025         DEBUG,
00026         INFO,
00027         WARNING,
00028         ERROR
00029     };
00030
00035     explicit Logger(LogLevel level = INFO) : minimumLevel(level) {}
00036
00042     void log(LogLevel level, const char* message) const;
00043
00044 private:
00045     LogLevel minimumLevel;
00046 };
00047
00048 #endif
```

## 6.5 main.cpp File Reference

```
#include <iostream>
#include "Logger.h"
#include "CalculationEngine.h"
```

### Functions

- int main ()

### 6.5.1 Function Documentation

#### 6.5.1.1 main()

```
int main ()
```

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